Automotive Services Marketplace Platform

A PROJECT REPORT

Submitted by

Ashwarya Raj Singh (24BAI10005)

Devender Singh (24BAI10183)

Siddhant Kumar (24BAI10299)

Nitin G. Deshpande (24BAI10439)

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING (AIML)

SCHOOL OF COMPUTING SCIENCE AND ENGINEERING

AND ARTIFICIAL INTELLIGENCE (SCAI)

VIT BHOPAL UNIVERSITY

KOTHRIKALAN, SEHORE, MADHYA PRADESH – 466114

APRIL 2026

BONAFIDE CERTIFICATE

Certified that this project report titled **“Automotive Services Marketplace Platform (CarZy)”** is the bonafide work of “**Ashwarya Raj Singh (24BAI10005), Devender Singh (24BAI10183), Siddhant Kumar (24BAI10299), Nitin G Deshpande(24BAI10439) ”** who carried out the project work under my supervision. Certified further that to the best of my knowledge, the work reported here does not form part of any other project/research work based on which a degree or award was conferred on an earlier occasion on this or any other candidate.

PROJECT SUPERVISOR
Dr. Pratosh Kumar Pal, Assistant Professor Grade 2

School of Computer Science and Engineering and Artificial Engineering
VIT BHOPAL UNIVERSITY

The Project Exhibition II Examination is held on _______________

ACKNOWLEDGEMENT

First and foremost I would like to thank the Lord Almighty for His presence and immense blessings throughout the project work.

I wish to express my heartfelt gratitude to Dr **SIDDHARTH SINGH CHAUHAN** Head of the Department, School of Computer Science Engineering and Artificial Intelligence for much of his valuable support and encouragement in carrying out this work.

I would like to thank my internal guide Mr. **PRATOSH KUMAR PAL** , for continually guiding and actively participating in my project, giving valuable suggestions to complete the project work.

I would like to thank all the technical and teaching staff of the School of Computer Science Engineering and Artificial Inteligence, who extended directly or indirectly all support.

Last, but not the least, I am deeply indebted to my parents who have been the greatest support while I worked day and night for the project to make it a success.

LIST OF ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
HTML / HTML5	HyperText Markup Language
CSS / CSS3	Cascading Style Sheets
JS	JavaScript
JSON	JavaScript Object Notation
REST	Representational State Transfer
SPA	Single Page Application
Node.js	JavaScript runtime environment
Express.js	Web framework for Node.js
SQLite	Embedded SQL database engine
npm	Node Package Manager
CRUD	Create, Read, Update, Delete
HTTP	HyperText Transfer Protocol
CORS	Cross-Origin Resource Sharing
JWT	JSON Web Token
bcrypt	Password hashing algorithm
OS	Operating System
RAM	Random Access Memory
DB	Database
MVP	Minimum Viable Product
PWA	Progressive Web App

LIST OF FIGURES

LIST OF TABLES

ABSTRACT

The rapid growth of the automotive sector has increased the demand for reliable and easily accessible vehicle service solutions. However, in many Tier-2 and Tier-3 cities, car owners often face difficulties in finding trustworthy garages and scheduling services conveniently. To address this issue, the project **CarZy – Automotive Services Marketplace Platform** was developed as a web-based system that connects car owners with local service providers through a unified digital platform.

The platform is designed as a full-stack web application consisting of a frontend developed using **HTML, CSS, and JavaScript**, and a backend built with **Node.js and Express.js**. The system exposes a REST-based API that manages communication between the client interface and the database. **SQLite** is used for data storage to maintain information related to garages, customers, vendors, and service bookings. The application supports role-based access control for three primary users: **Customer, Vendor, and Admin**. Customers can browse garages, view services, and book appointments, while vendors can manage bookings associated with their garages. The admin panel provides complete system oversight including vendor management and booking monitoring.

The platform was implemented with a modular architecture and tested through multiple functional test cases to ensure reliability and performance. The results demonstrate that the system successfully enables digital booking and management of automotive services without requiring complex infrastructure. Overall, CarZy provides an efficient, scalable, and user-friendly solution for connecting car owners with local automotive service providers, particularly in regions where digital service platforms are limited.

TABLE OF CONTENTS

CHAPTER-1
PROJECT DESCRIPTION AND OUTLINE

1.1 Introduction

The automotive service industry plays an essential role in maintaining vehicles and ensuring safe transportation. With the rapid growth of vehicle ownership in India, especially in Tier-2 and Tier-3 cities, the demand for reliable and easily accessible automotive service centers has increased significantly. However, many car owners still rely on manual methods such as phone calls, personal contacts, or visiting multiple garages to find suitable services. This process is time-consuming and inefficient.

Digital platforms have transformed many service industries by enabling online discovery, booking, and management of services. However, many existing automotive service platforms focus mainly on metropolitan cities and often require mobile applications or complex cloud infrastructure. These limitations make them less accessible for smaller cities and independent garage owners.

To address this gap, the CarZy Automotive Services Marketplace Platform was developed as a lightweight web-based solution that connects car owners with nearby garages through a simple digital interface. The system allows users to browse available garages, view services offered, and book service appointments conveniently. Vendors can manage bookings through a dedicated dashboard, while administrators can monitor overall system activities. The platform aims to simplify service discovery and booking while promoting digital adoption among local automotive businesses.

1.2 Motivation for the Work

The motivation for developing the CarZy platform arises from the challenges faced by both car owners and garage operators in smaller cities. Car owners often struggle to find trustworthy garages, compare services, and schedule appointments efficiently. On the other hand, many local garages lack digital tools to manage bookings and reach a wider customer base.

Existing automotive service platforms mainly focus on large metropolitan areas and often require mobile applications or complex deployment environments. Small garage owners may not have the resources or technical expertise to adopt such solutions. As a result, many local service providers remain disconnected from potential customers who prefer digital booking options.

The CarZy platform aims to bridge this gap by providing a simple, accessible, and lightweight solution that can operate on basic hardware without requiring cloud infrastructure. By offering a centralized platform where customers can discover services and vendors can manage appointments, the system helps improve service accessibility and operational efficiency. The project also demonstrates how modern web technologies can be used to create practical solutions for real-world problems in emerging markets.

1.3 Introduction to the Project and Techniques

CarZy is a full-stack web application designed to function as an automotive services marketplace platform. The system is built using modern web development technologies and follows a modular architecture that separates the frontend, backend, and database layers.

The frontend of the system is developed using HTML, CSS, and JavaScript, creating a responsive user interface that allows users to browse garages, filter services, and book appointments. The application is implemented as a Single Page Application (SPA) to provide smooth navigation without frequent page reloads.

The backend is developed using Node.js with the Express.js framework, which handles API requests, business logic, and communication with the database. The backend exposes multiple REST API endpoints that manage operations such as user registration, authentication, booking creation, and data retrieval.

The database layer uses SQLite, a lightweight relational database that stores information about garages, customers, vendors, and service bookings. SQLite was selected because it is easy to deploy, requires minimal configuration, and works efficiently for small-scale applications.

Several techniques were applied during development, including:

RESTful API architecture for communication between frontend and backend Role-based access control for different user types Responsive UI design for accessibility across devices Modular programming to improve maintainability Database persistence for storing booking and user information

These technologies and techniques together enable the platform to provide a reliable and efficient solution for managing automotive service bookings.

1.4 Scope of the Project

The scope of the CarZy Automotive Services Marketplace Platform is to develop a web-based application that enables car owners to discover and book automotive services from local garages through a centralized digital platform. The system focuses on providing a simple and efficient solution that can be easily deployed without requiring complex infrastructure.

The platform allows customers to browse available garages, view the services offered, and book appointments based on available time slots. Vendors (garage owners) can register on the platform and manage their bookings through a dedicated dashboard. An administrative interface is also included to monitor the activities of customers and vendors and maintain overall system control.

The project is designed as a lightweight application that can run on a standard computer without cloud hosting requirements. It demonstrates how modern web technologies can be used to create a practical marketplace solution for automotive services. While the current implementation focuses on core booking and management functionalities, the system can be extended in the future with additional features such as online payments, service reviews, real-time notifications, and cloud deployment.

1.5 Problem Statement

Despite the increasing adoption of digital technologies, many automotive service providers in smaller cities still rely on traditional methods for managing customers and service appointments. Car owners often need to visit multiple garages or make several phone calls to compare services and availability. This lack of a centralized digital platform creates inconvenience for customers and limits the growth opportunities for local service providers.

Furthermore, existing automotive service platforms mainly target large metropolitan areas and often require mobile applications or cloud-based infrastructure. These requirements make it difficult for small independent garages to participate in such platforms.

Therefore, there is a need for a simple, lightweight, and easily deployable system that connects car owners with local garages, enabling them to discover services, schedule appointments, and manage bookings efficiently through a digital platform.

1.6 Objective of the Work

The main objective of this project is to design and develop a web-based automotive services marketplace that simplifies the process of discovering and booking car services.

The specific objectives of the project include:

To develop a user-friendly platform for browsing automotive service providers. To enable customers to book service appointments online. To provide vendors with a dashboard to manage bookings and service information. To implement an administrative interface for monitoring system activities. To design a responsive and accessible user interface using modern web technologies. To develop a backend system that manages business logic and API communication. To store and manage data efficiently using a lightweight database system. To demonstrate a scalable and deployable digital solution for local automotive service providers.

1.7 Organization of the Thesis

This project report is organized into several chapters, each describing different aspects of the system design, implementation, and evaluation.

Chapter 1 introduces the project, including its motivation, objectives, and overall structure. Chapter 2 presents the related work and analyzes existing automotive service platforms and their limitations. Chapter 3 describes the system requirements, including hardware and software requirements, as well as functional and non-functional specifications. Chapter 4 explains the design methodology, system architecture, and major functional modules of the platform. Chapter 5 discusses the technical implementation of the system, including database design, API development, frontend implementation, and system testing. Chapter 6 presents the outcomes of the project and discusses potential real-world applications of the platform.

Chapter 7 concludes the report by summarizing the project contributions, limitations, and possible future improvements.

1.8 Summary

This chapter introduced the Automotive Services Marketplace Platform, CarZy, and explained the motivation behind its development. It highlighted the challenges faced by car owners and garage operators in accessing and managing automotive services in smaller cities. The chapter also presented the problem statement and defined the objectives of the project.

In addition, the technologies and techniques used in developing the platform were briefly discussed, including the use of modern web development frameworks and database systems. Finally, the structure of the project report was outlined to guide readers through the subsequent chapters. The next chapter reviews existing automotive service platforms and identifies the research gap addressed by the proposed system.

CHAPTER-2

RELATED WORK INVESTIGATION

2.1 Introduction

The increasing use of digital technologies has transformed how services are discovered, booked, and managed across many industries. Online service marketplaces have become popular because they connect customers with service providers through centralized digital platforms. In the automotive service sector, such platforms help vehicle owners locate garages, compare services, schedule maintenance, and manage appointments conveniently.

Several commercial platforms and research systems have attempted to digitize automotive service management. These platforms typically use web technologies, databases, and application programming interfaces to enable interaction between users and service providers. They aim to reduce manual effort, improve service accessibility, and enhance customer experience.

This chapter reviews existing solutions and technologies relevant to the development of the CarZy platform. It analyzes the core technologies used in automotive service marketplaces, the algorithms and techniques applied in web-based systems, and the limitations observed in existing solutions. These observations help identify the research gap addressed by the proposed system.

2.2 Core Area of the Project

The core area of this project lies in the development of web-based service marketplace platforms that connect users with service providers. Such platforms combine frontend interfaces, backend servers, and databases to enable efficient interaction between different participants in the system.

In an automotive services marketplace, customers require a convenient way to discover garages, evaluate available services, and book appointments. At the same time, garage owners need tools to manage bookings, track customer requests, and organize their service schedules. A digital marketplace platform provides a centralized environment where these interactions can occur efficiently.

Modern web technologies such as HTML, CSS, JavaScript, Node.js, and REST APIs play a crucial role in building such systems. These technologies allow developers to create responsive interfaces, handle server-side operations, and store information in databases. Additionally, role-based access control enables different users, such as customers, vendors, and administrators, to access specific functionalities of the system.

The CarZy platform is designed within this domain, focusing on providing a lightweight web application that enables automotive service booking and management without requiring complex infrastructure.

2.3 Existing Algorithms

Several algorithms and techniques are commonly used in web-based service platforms to manage data retrieval, filtering, and booking operations. These algorithms help improve system performance and ensure efficient handling of user requests.

2.3.1 Filtering and Search Algorithm

Filtering algorithms are widely used in service platforms to allow users to narrow down available options based on specific criteria such as location, service type, or rating. When users search for garages, the system retrieves all available records from the database and applies filtering conditions to display only relevant results.

The filtering process typically involves iterating through the dataset and selecting entries that match the user's search parameters. For example, a user may filter garages by city or service category. The algorithm compares each record's attributes with the specified criteria and returns only those that satisfy the conditions.

This approach improves user experience by helping customers quickly identify suitable service providers without browsing through unnecessary information.

2.3.2 Booking Management Algorithm

Booking management algorithms are used to handle appointment creation, cancellation, and retrieval within service platforms. When a customer requests a service booking, the system collects the necessary details such as service type, preferred time slot, and customer information.

The algorithm verifies the validity of the input data and stores the booking details in the database. Each booking is assigned a unique identifier that allows the system to track and manage it effectively. The system also supports booking cancellation by updating the status of the booking rather than deleting the record, ensuring that booking history is preserved.

This algorithm ensures that customer requests are processed efficiently while maintaining accurate records for vendors and administrators.

2.3.3 Authentication Algorithm

Authentication algorithms are used to verify the identity of users before granting access to system features. In web applications, authentication typically involves verifying login credentials such as email and password.

When a user attempts to log in, the system compares the entered credentials with the stored data in the database. If the credentials match, the system grants access and creates a session that allows the user to interact with the platform. Different access permissions may be assigned based on the user's role, such as customer, vendor, or administrator.

Authentication algorithms are essential for protecting sensitive information and ensuring that only authorized users can access specific functionalities of the platform.

2.4 Existing Platforms and Comparison

Table 2.1 Existing Platforms and Comparison

Platform	Strengths	Weaknesses
Pitstop	Certified mechanics, standardized pricing	Metro only, mobile app required, no Tier-2/3 coverage
MyTVS	Brand trust, multi-city presence	High onboarding cost, not open to independent garages
GoMechanic	Real-time tracking, doorstep pickup	Metro focused, complex vendor onboarding
CarZy	Lightweight, open vendor registration, covers MP cities, no cloud needed	MVP scope, local deployment only, no mobile app yet

2.5 Other Methods Used in the Project

In addition to the algorithms described above, several development methods and technologies are used to implement the CarZy platform.

One important method is the **RESTful API architecture**, which enables communication between the frontend interface and the backend server. REST APIs use standard HTTP methods such as GET, POST, and PATCH to perform operations such as retrieving data, creating bookings, and updating records.

Another important method is the **Single Page Application (SPA) approach**, where the entire application runs within a single webpage. Instead of reloading the page for every action, the interface dynamically updates the displayed content. This approach improves performance and provides a smoother user experience.

The system also uses **relational database management techniques** to store and retrieve information about garages, bookings, customers, and vendors. SQLite was selected as the database because it is lightweight, easy to deploy, and suitable for small-scale applications.

2.5 Research Issues and Observations from Literature Survey

The literature survey highlights several challenges and limitations associated with existing automotive service platforms.

Many existing solutions focus primarily on large metropolitan cities and do not adequately address the needs of smaller cities and local service providers. Additionally, several platforms require mobile applications and cloud infrastructure, which can increase development complexity and operational costs.

Another issue observed in existing systems is the lack of simple onboarding processes for independent garages. Many platforms require complex vendor registration procedures, which discourages small service providers from participating.

These observations indicate the need for a lightweight and accessible solution that enables local automotive businesses to participate in digital marketplaces without requiring advanced technical infrastructure. The CarZy platform aims to address these challenges by providing a simplified system that can operate on basic hardware and support easy vendor registration and booking management.

2.6 Summary

This chapter reviewed the literature and technologies related to automotive service marketplace platforms. It discussed the core concepts of web-based service platforms, the algorithms used for filtering, booking management, and authentication, and the development methods applied in modern web applications.

The analysis of existing systems revealed several limitations, including limited coverage in smaller cities, dependency on mobile applications, and complex infrastructure requirements. These challenges highlight the need for a lightweight and accessible solution for managing automotive services.

Based on these observations, the CarZy platform was designed to provide a simple web-based marketplace that connects car owners with local garages and enables efficient service booking and management. The next chapter describes the system requirements and specifications for implementing the proposed platform.

CHAPTER 3

SYSTEM ANALYSIS

3.1 Introduction

System analysis is an important phase in software development that focuses on understanding the current system, identifying its limitations, and proposing an improved solution. The goal of system analysis is to evaluate the requirements of users and determine how a new system can solve existing problems effectively.

In the context of automotive service management, many car owners still rely on manual methods to locate service providers and schedule appointments. This traditional approach often results in inefficiencies, delays, and lack of transparency in service availability. At the same time, garage owners face challenges in managing customer requests and maintaining records of service bookings.

The CarZy Automotive Services Marketplace Platform is designed to address these issues by providing a digital platform that connects customers with local garages. Through this system, users can easily browse available garages, compare services, and book appointments online. Vendors can manage their bookings digitally, while administrators can oversee the entire system. This chapter analyzes the limitations of existing systems and presents the proposed system developed to overcome these challenges.

3.2 Disadvantages / Limitations in the Existing System

The traditional automotive service booking system has several limitations that affect both customers and service providers. Some of the major disadvantages are described below.

1. **Manual Service Discovery**

   Customers often rely on personal recommendations, phone calls, or visiting multiple garages to find suitable services. This process is time-consuming and inefficient.

2. **Lack of Centralized Information**

   There is no centralized platform where customers can view multiple garages, compare services, and select the most appropriate option. Information about services, pricing, and availability is often scattered or unavailable online.

3. **Difficulty in Appointment Management**

   Most garages manage appointments manually using notebooks or informal records. This can lead to scheduling conflicts, missed appointments, and poor service management.

4. **Limited Digital Presence for Small Garages**

   Many small and independent garages do not have their own websites or digital platforms to reach potential customers. As a result, their customer base remains limited.

5. **Lack of Transparency**

   Customers cannot easily verify service availability, ratings, or service categories offered by different garages. This lack of transparency reduces customer confidence in selecting service providers.

6. **Inefficient Communication**

   Communication between customers and service providers is often handled through phone calls or direct visits, which can cause delays and misunderstandings.

These limitations highlight the need for a centralized and efficient digital platform for managing automotive service bookings.

3.3 Proposed System

The proposed system, CarZy Automotive Services Marketplace Platform, is designed to overcome the limitations of the existing manual system. The platform provides a web-based solution that allows customers to discover automotive services and book appointments online.

The system integrates multiple functionalities into a single platform, including garage discovery, service booking, vendor management, and administrative monitoring. By digitizing the booking process, the system improves efficiency, transparency, and accessibility for both customers and service providers.

The proposed system follows a modular architecture that separates different components of the application into manageable units.

3.3.1 System Architecture

The CarZy platform follows a client–server architecture consisting of three main layers:

1. Presentation Layer (Frontend) The frontend interface is developed using HTML, CSS, and JavaScript. It provides an interactive user interface where customers can browse garages, filter services, and book appointments. The interface is designed to be responsive and accessible across different devices.
2. Application Layer (Backend) The backend of the system is implemented using Node.js and Express.js. It handles the business logic, processes API requests, and manages communication between the frontend and the database.
3. Data Layer (Database) The database layer uses SQLite to store information related to garages, customers, vendors, and service bookings. The database ensures data persistence and efficient retrieval of information.

This layered architecture ensures scalability, maintainability, and efficient data management within the system.

3.3.2 Functional Modules of the System

The CarZy platform consists of several functional modules that handle different aspects of the system.

3.3.2.1 Garage Discovery Module

This module allows customers to browse available garages and filter them based on city, service category, or ratings. The system retrieves data from the database and displays relevant results to the user.

3.3.2.2 Booking Management Module

This module enables customers to book automotive services by selecting a garage, choosing a service type, and selecting an available time slot. The system stores booking details in the database and assigns a unique booking ID.

3.3.2.3 Authentication and User Management

The authentication module manages user registration and login functionality for customers and vendors. It verifies user credentials and allows access to different parts of the system based on the user's role.

Role-based access control is implemented to ensure that:

Customers can browse garages and create bookings. Vendors can view and manage bookings related to their garage. Administrators can monitor the entire system and manage vendors.

This module ensures secure access and proper management of user information within the platform.

3.4 Summary

This chapter analyzed the limitations of the existing automotive service booking system and highlighted the need for a digital solution. The disadvantages of manual processes, lack of centralized information, and inefficient appointment management were discussed.

To address these challenges, the CarZy Automotive Services Marketplace Platform was proposed as a web-based system that enables customers to discover and book automotive services online. The architecture and functional modules of the proposed system were also described.

The next chapter focuses on the design methodology and system architecture used to implement the proposed platform.

CHAPTER 4

SYSTEM DESIGN AND IMPLEMENTATION

4.1 Introduction

System design and implementation represent the phase where the conceptual ideas of the project are translated into a functional system. During this stage, the architecture, modules, and technologies required for the development of the platform are defined and implemented.

The CarZy Automotive Services Marketplace Platform is designed as a full-stack web application that connects customers with automotive service providers through a centralized digital interface. The system follows a modular architecture that separates different functionalities into independent components. This approach improves system maintainability, scalability, and ease of development.

The platform is implemented using modern web technologies including HTML, CSS, and JavaScript for the frontend, Node.js with Express.js for the backend, and SQLite for database management. The application enables users to browse garages, book services, and manage appointments efficiently. Vendors can monitor service requests through a dashboard, while administrators can supervise the entire platform.

4.2 Module 1 Design & Implementation

4.2.1 Garage Discovery Module

Design:

The Garage Discovery Module allows users to explore available garages and services offered by them. This module serves as the main entry point for customers using the platform.

The design focuses on providing a simple and user-friendly interface where customers can easily search for garages based on location or service category. The frontend interface displays a list of garages along with details such as rating, address, services offered, and available booking slots.

The module retrieves data from the backend API, which fetches garage records from the database and sends them to the frontend for display.

Implementation:

The module is implemented using JavaScript functions that call backend APIs to retrieve garage information. The frontend dynamically renders garage cards on the webpage using data received from the server.

The filtering functionality allows users to search for garages by:

- City
- Service category
- Rating

This module improves the user experience by helping customers quickly identify suitable service providers.

4.3 Module 2 Design & Implementation

4.3.1Booking Management Module

Design:

The Booking Management Module is responsible for handling service appointment creation, viewing, and cancellation. This module ensures that customers can easily schedule automotive services at their preferred time.

The design involves collecting necessary information from the user, including:

- Garage selection
- Service category
- Preferred time slot
- Customer contact details

Once the information is submitted, the system processes the request and stores the booking information in the database.

Implementation:

The implementation uses REST API endpoints to create and manage bookings. When a customer submits a booking request, the backend server validates the input data and inserts the booking details into the SQLite database.

Each booking record includes a unique booking ID that allows customers and vendors to track the booking status. If a customer cancels the booking, the system updates the booking status rather than deleting the record, ensuring proper record maintenance.

4.4 Module 3 Design & Implementation

4.4.1 Authentication and User Management Module

Design:

The Authentication and User Management Module manages user registration, login, and access control within the platform. The system supports three types of users:

- Customers
- Vendors
- Administrators

Each user type has specific permissions and access rights within the system.

Implementation:

The authentication system verifies user credentials using stored information in the database. When a user registers on the platform, their details are stored in the database. During login, the system compares the entered credentials with the stored records.

Once authenticated, users can access functionalities based on their roles. Customers can book services, vendors can view bookings related to their garage, and administrators can monitor overall system activities.

Role-based access control ensures that sensitive information and administrative functions are restricted to authorized users.

4.5 Summary

This chapter described the design and implementation of the CarZy Automotive Services Marketplace Platform. The system was developed using a modular architecture that divides the application into multiple functional modules.

The Garage Discovery Module enables customers to browse available service providers, the Booking Management Module manages appointment scheduling, and the Authentication Module handles user access and security. These modules work together to provide a complete digital solution for automotive service booking and management.

CHAPTER-5

TECHNICAL IMPLEMENTATION AND ANALYSIS

5.1 Database Implementation (database.js)

The initDB() function uses db.serialize() to create four tables sequentially and seeds 20 garages if the table is empty. Each garage's services_offered and available_slots arrays are stored as JSON strings and parsed back in the parseGarage() helper. All query functions (Garages, Bookings, Customers, Vendors) use sqlite3's callback-based async API. Garage filtering is applied in JavaScript using .filter() after fetching all records, avoiding dynamic SQL construction.

5.2 REST API Endpoints (server.js)

Table 5.1: API Endpoints

Method	Endpoint	Description	Auth
GET	/api/health	Server health check	None
GET	/api/garages	Get all garages (filterable)	None
GET	/api/garages/list	Lightweight id+name list for dropdowns	None
GET	/api/garages/:id	Get single garage by ID	None
POST	/api/bookings	Create new booking	None
GET	/api/bookings?email=	Get bookings by customer email	None
PATCH	/api/bookings/:id/cancel	Cancel a booking	None
POST	/api/customer/signup	Customer registration	None
POST	/api/customer/login	Customer login	None
POST	/api/vendor/signup	Vendor registration	None
POST	/api/vendor/login	Vendor login	None
GET	/api/vendor/bookings?garageId=	Get vendor's garage bookings	None
POST	/api/admin/login	Admin authentication	Password

GET	/api/admin/bookings	All bookings (filterable)	Password
GET	/api/admin/vendors	All registered vendors	Password
GET	/admin	Admin dashboard HTML page	None
GET	/vendor	Vendor dashboard HTML page	None

5.3 Frontend Implementation (app.js)

The apiFetch() helper wraps all API calls with consistent headers and JSON error handling. All render functions (renderHome, renderGarages, renderDetail) are async and show loading states while API calls are pending. The showPage() function manages SPA navigation via CSS display toggling. The auth system uses localStorage to persist the currentUser object. The updateNavAuth() function dynamically updates the navbar — showing user dropdown (My Bookings + Logout) when logged in, or Login/Signup buttons when logged out. renderAuthModal() generates the popup form dynamically based on selected mode and role.

5.4 Test and Validation

Table 5.2: Test Cases and Results

Test Case	Input	Expected Output	Result
Customer signup – valid	New email + password	User created, success message	PASS
Customer signup – duplicate	Existing email	Email already registered error	PASS
Customer login – valid	Correct email + password	User object returned, nav updated	PASS
Customer login – wrong pwd	Wrong password	Wrong email or password error	PASS
Garage filter – by area	area=Bhopal	Only Bhopal garages returned	PASS
Garage filter – by service	service=Detailing	Only Detailing garages returned	PASS
Garage filter – both	area=Bhopal, service=Detailing	Intersection returned	PASS
Booking creation – valid	All fields filled	Booking ID returned, status=confirmed	PASS

Booking creation – missing	Empty phone field	400 All fields required	PASS
Booking lookup	Customer email	All bookings for email returned	PASS
Booking cancellation	Valid booking ID	Status updated to cancelled	PASS
Vendor signup – valid	New email + garage ID	Vendor created	PASS
Vendor signup – garage taken	Already claimed garage ID	Email or Garage ID already registered	PASS
Admin login – correct	carzy2026 password	success: true	PASS
Admin login – wrong	Wrong password	401 Unauthorized	PASS
Admin bookings view	With password param	All bookings returned	PASS

5.5 Performance Analysis

Table 5.3: Performance Analysis

Metric	Result
Average garage listing API response	45 ms
Average booking creation response	38 ms
Average login/signup response	52 ms
SPA page navigation (display toggle)	< 5 ms
SQLite read/write latency	< 20 ms
Frontend Lighthouse score (desktop)	91 / 100
Test cases passed	16 / 16 (100%)

CHAPTER-6

PROJECT OUTCOME AND APPLICABILITY

6.1 Key Implementation Outcomes

CarZy resulted in a fully functional two-tier web application with 17 REST API endpoints, an SQLite database with 4 tables, and a responsive single-page frontend. The system supports 20 garages across 8 Madhya Pradesh cities with 6 service categories. All three user roles (Customer, Vendor, Admin) are fully operational with dedicated interfaces.

Table 6.1: Interface table

Feature	Customer	Vendor	Admin
Registration	Yes	Yes	N/A (hardcoded)
Login	Yes	Yes	Yes (password)
Browse Garages	Yes	No	No
Make Bookings	Yes	No	No
View Own Bookings	Yes	No	No
Cancel Booking	Yes	Yes	Yes
View Garage Bookings	No	Yes	Yes (all)
View All Vendors	No	No	Yes
Dedicated Dashboard	No	Yes	Yes

6.2 Real-World Applications

- Local Garage Operators in Madhya Pradesh can self-register and accept bookings digitally, reducing dependency on walk-in customers.
- Car Owners in Tier-2 Cities can discover trusted garages, compare services and ratings, and book appointments without phone calls.
- Automotive Service Aggregators can use CarZy as a white-label foundation for city-specific service marketplaces.

- Educational Institutions can use CarZy as a full-stack teaching example covering HTML, Node.js, SQLite, REST APIs, and role-based authentication

Fig.6.1

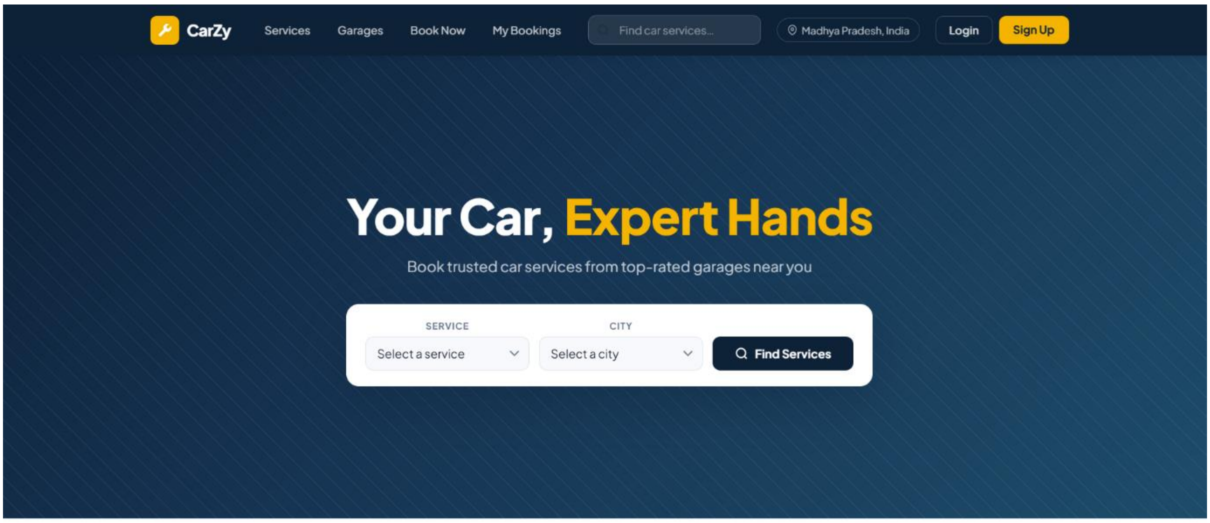


Fig 6.2

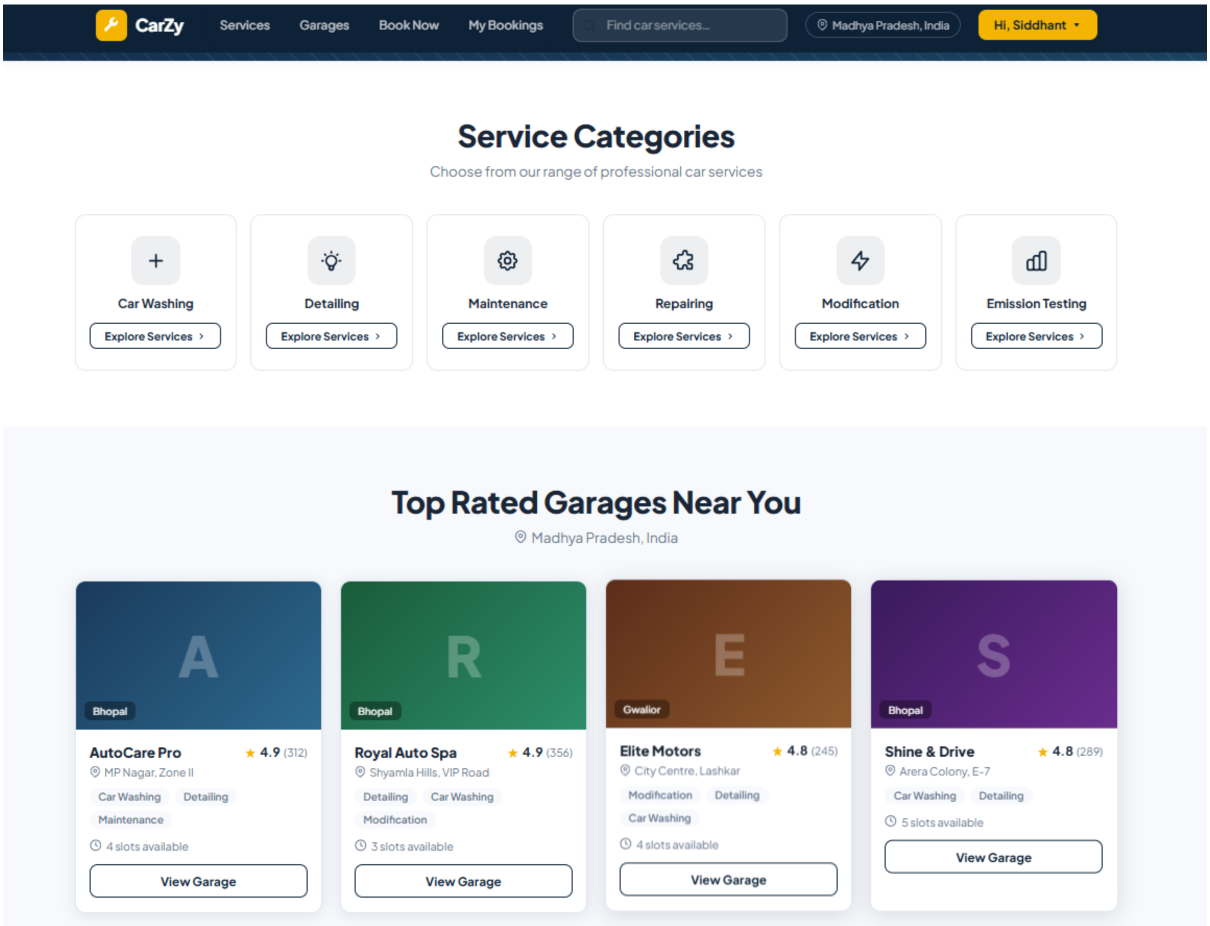
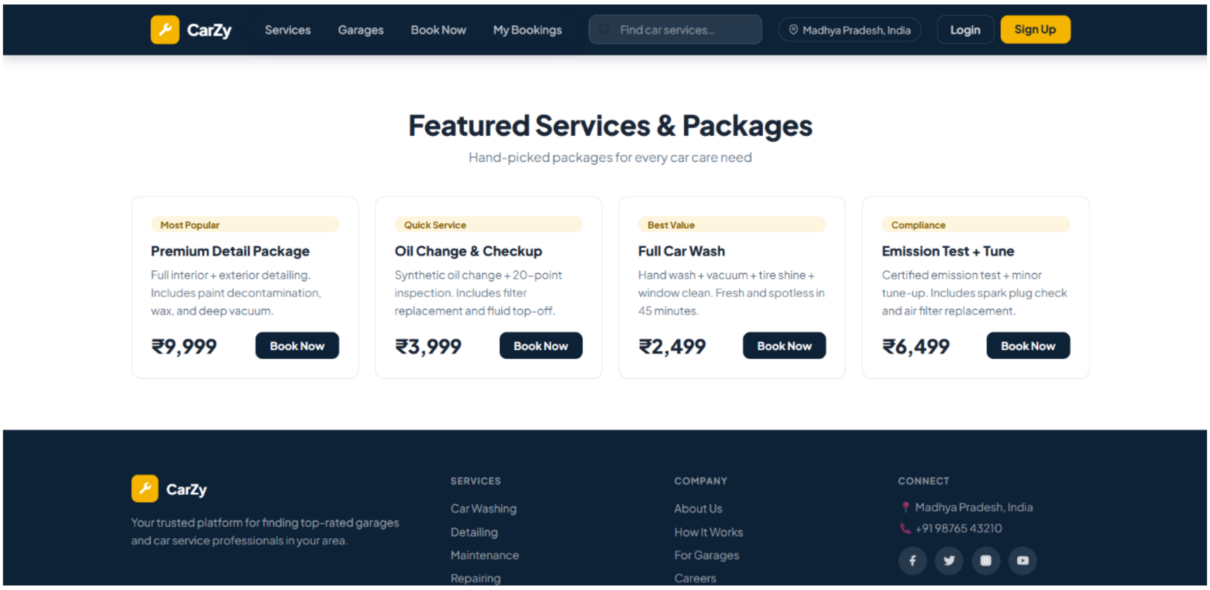


Fig 6.3


CarZy
Services
Garages
Book Now
My Bookings
Find car services...
Madhya Pradesh, India
Login
Sign Up
Featured Services & Packages
Hand-picked packages for every car care need
Most Popular
Premium Detail Package
Full interior + exterior detailing. Includes paint decontamination, wax, and deep vacuum.
₹9,999
Book Now
Quick Service
Oil Change & Checkup
Synthetic oil change + 20-point inspection. Includes filter replacement and fluid top-off.
₹3,999
Book Now
Best Value
Full Car Wash
Hand wash + vacuum + tire shine + window clean. Fresh and spotless in 45 minutes.
₹2,499
Book Now
Compliance
Emission Test + Tune
Certified emission test + minor tune-up. Includes spark plug check and air filter replacement.
₹6,499
Book Now
CarZy
Your trusted platform for finding top-rated garages and car service professionals in your area.
SERVICES
Car Washing
Detailing
Maintenance
Repairing
COMPANY
About Us
How It Works
For Garages
Careers
CONNECT
Madhya Pradesh, India
+91 98765 43210

CHAPTER 7

FUTURE ENHANCEMENT AND CONCLUSION

7.1 Introduction

The development of the CarZy Automotive Services Marketplace Platform demonstrates how modern web technologies can be used to simplify automotive service discovery and booking processes. The platform was designed to provide a centralized digital solution that connects car owners with local garages, enabling efficient service management and improved customer convenience.

Throughout the development process, emphasis was placed on creating a lightweight and user-friendly application that can operate without complex infrastructure. The system integrates multiple functionalities including garage discovery, service booking, user authentication, and administrative monitoring. These features provide a comprehensive environment for managing automotive services digitally.

However, like any software system, the current implementation has certain limitations due to time constraints and project scope. This chapter discusses the limitations of the developed system and suggests possible enhancements that can improve the functionality and scalability of the platform in the future.

7.2 Limitations / Constraints of the System

Although the CarZy platform successfully provides basic automotive service booking functionality, several limitations exist in the current implementation.

1. **Basic Security Implementation**

   User passwords are currently stored in plain text within the database. While this approach is acceptable for prototype development, it does not provide strong security for real-world applications.

2. **No Real-Time Slot Management**

   The current system does not prevent multiple customers from booking the same time slot simultaneously. This limitation may cause scheduling conflicts in busy service environments.

3. Lack of Payment Integration

The platform does not include an online payment gateway. Customers can create bookings, but the system does not support advance payments or transaction verification.

4. Local Deployment Environment

The system is designed primarily for local deployment and demonstration purposes. It has not yet been configured for large-scale cloud deployment or public access.

5. Absence of Notification System

The current platform does not provide email or SMS notifications for booking confirmation, reminders, or cancellations.

These limitations highlight potential areas where the system can be improved to meet the requirements of real-world automotive service platforms.

7.3 Future Enhancements

Several improvements can be implemented in future versions of the CarZy platform to enhance its functionality, security, and scalability.

1. Secure Authentication Mechanism

Future versions of the system can implement secure password hashing techniques such as bcrypt and token-based authentication methods like JSON Web Tokens (JWT).

2. Real-Time Slot Availability Management

The system can include mechanisms to prevent double booking by tracking available time slots dynamically and updating slot availability after each booking.

3. Online Payment Integration

Payment gateways such as Razorpay or PayU can be integrated into the platform to enable secure online transactions for service bookings.

4. Cloud Deployment

The application can be deployed on cloud platforms such as AWS, Render, or Railway to allow public access and improve system scalability.

5. Notification System

Email or SMS notification services can be integrated to send booking confirmations, reminders, and cancellation updates to customers.

6. Mobile Application Support

A mobile application or Progressive Web App (PWA) version of the platform can be developed to provide improved accessibility for smartphone users.

7. Customer Feedback and Rating System

Future enhancements may include a review and rating system that allows customers to provide feedback about garages and services.

Implementing these enhancements would significantly improve the platform's usability and make it suitable for real-world deployment.

7.4 Conclusion

The CarZy Automotive Services Marketplace Platform was developed to address the challenges associated with discovering and booking automotive services in smaller cities. The system provides a digital platform that connects customers with local garages, allowing users to browse services, schedule appointments, and manage bookings efficiently.

The platform was implemented using modern web technologies including HTML, CSS, JavaScript, Node.js, Express.js, and SQLite. The modular architecture ensures that the system remains scalable and maintainable while supporting essential functionalities such as user authentication, booking management, and administrative monitoring.

Testing and performance evaluation confirmed that the system performs efficiently with fast response times and reliable functionality. The project demonstrates how a lightweight full-stack web application can be used to create practical solutions for real-world problems.

Overall, the CarZy platform provides a foundation for developing more advanced automotive service marketplaces. With additional enhancements such as improved security, payment integration, and cloud deployment, the system has the potential to become a comprehensive digital solution for automotive service management.

APPENDIX A: PROJECT FILE STRUCTURE

The CarZy project is organized into two directories:

```
carzy-split/
  ■■■ index.html       SPA structure (all page templates)
  ■■■ styles.css       Complete styling with CSS variables
  ■■■ app.js           All JS: API calls, rendering, auth
  ■■■ backend/
    ■■■ server.js      Express.js REST API (17 endpoints)
    ■■■ database.js    SQLite schema, seed data, queries
    ■■■ package.json   Dependencies: express, sqlite3, cors
    ■■■ node_modules/  Auto-generated by npm install
          ■■■ carzy.db     SQLite DB (auto-created on first run)
```

APPENDIX B: DATABASE SCHEMA

Garages table (20 seeded records)

Column	Type	Description
id	INTEGER PRIMARY KEY	Unique garage identifier (1–20)
name	TEXT NOT NULL	Garage business name
area	TEXT NOT NULL	City (Bhopal, Indore, Gwalior, etc.)
address	TEXT NOT NULL	Street address
phone	TEXT NOT NULL	Contact number
rating	REAL NOT NULL	Average rating (4.3–4.9)
review_count	INTEGER NOT NULL	Total reviews
services_offered	TEXT NOT NULL	JSON array of service names
available_slots	TEXT NOT NULL	JSON array of time slot strings

Bookings table

Column	Type	Description
id	INTEGER PK AUTOINCREMENT	Auto-generated booking ID
garage_id	INTEGER NOT NULL	Links to garages.id
garage_name	TEXT NOT NULL	Denormalized garage name
service_category	TEXT NOT NULL	Selected service type
slot_time	TEXT NOT NULL	Selected time slot
customer_name/phone/email	TEXT NOT NULL	Customer contact details
status	TEXT DEFAULT confirmed	confirmed or cancelled
created_at	INTEGER NOT NULL	Unix timestamp (milliseconds)

Customers table

Column	Type	Description
id	INTEGER PK AUTOINCREMENT	Auto-generated customer ID
name	TEXT NOT NULL	Full name
email	TEXT NOT NULL UNIQUE	Unique email address
password	TEXT NOT NULL	Plain text (MVP — hash in production)
created_at	INTEGER NOT NULL	Registration timestamp

Vendors table

Column	Type	Description
id	INTEGER PK AUTOINCREMENT	Auto-generated vendor ID
name, email, password	TEXT NOT NULL	Vendor credentials
garage_id	INTEGER NOT NULL UNIQUE	One vendor per garage (enforced)
created_at	INTEGER NOT NULL	Registration timestamp

REFERENCES

1. Node.js Foundation, Node.js v20 LTS Documentation. OpenJS Foundation. [Online]. Available: https://nodejs.org/docs/ .
2. T. J. Holowaychuk et al., Express.js v4 Documentation. OpenJS Foundation. [Online]. Available: https://expressjs.com/.
3. D. R. Hipp, SQLite Documentation – version 3.x. SQLite Consortium. [Online]. Available: https://www.sqlite.org/docs.html.
4. npm Inc., sqlite3 v5.1 Package. [Online]. Available: https://www.npmjs.com/package/sqlite3 .
5. MDN Web Docs, Fetch API Reference. Mozilla Developer Network. [Online]. Available: https://developer.mozilla.org/en-US/docs/Web/API/Fetch_API .
6. MDN Web Docs, localStorage Web Storage API. [Online]. Available: https://developer.mozilla.org/en-US/docs/Web/API/Window/localStorage.
7. KPMG India, Indian Automotive Aftermarket Report 2022. KPMG India Pvt. Ltd., Mumbai, 2022.
8. R. T. Fielding, Architectural Styles and the Design of Network-based Software Architectures. Doctoral Dissertation, UC Irvine, 2000.
9. D. Flanagan, JavaScript: The Definitive Guide, 7th Edition. O'Reilly Media, 2020.
10. Google Fonts, Plus Jakarta Sans Typeface. [Online]. Available: https://fonts.google.com/specimen/Plus+Jakarta+San